补充资料 / Supplementary methods, diagnostics and event schedules

本补充资料与 2026 年 10 月 23 日中英文稿对应。参数校准与 Sobol 沿用已核验来源；优化主体采用 2026 年 10 月 22 日完成的五期 30 维搜索。来源版本、旧控制空间及条件结构试验分别记录。

This supplement accompanies both updated manuscripts. Calibration and Sobol data retain their verified sources. Main optimization uses the five-stage, 30-dimensional searches completed on 22 October 2026. Earlier control spaces and conditional structural experiments are recorded separately.

S1 数据来源与执行日历 / Input provenance and model calendar

Table S1. Newly supplied data compared with frozen model inputs.

Year	Weather rows	Max weather difference	RS rows	RS label
2019	365	0.0	23	BI
2020	366	0.0	23	BI
2021	365	0.0	22	BI
2022	365	0.0	17	not supplied

Note. The two supplied 2021 remote-sensing files are byte-identical. Remote-sensing inputs remain auxiliary; calibration uses DI LAI-derived CC and digitized biomass.

Water totals use actual executed daily dates. The annual scripts retain their end-boundary conventions; phenological date eligibility is the intersection listed in Table S4. Raw reference weather and model-used audited ETo are separately retained.

S2 校准先验 / Calibration priors

Table S2. Recorded annual phase-specific priors.

Year	Phase	Parameter	Family	Support	Center	Scale
2019	1	CCx	Uniform	0.6–0.99	–	–
2019	1	CGC_CD	Uniform	0.08–0.16	–	–
2019	1	CDC_CD	Truncated normal	0.03–0.12	0.07	0.025
2019	1	SenescenceCD	Truncated normal	115–145	132	10
2019	1	Kcb	Uniform	1.05–1.6	–	–
2019	2	WP_early	Uniform	10–40	–	–
2019	2	WP_late	Uniform	10–40	–	–
2019	2	HI0	Uniform	0.35–0.6	–	–
2020	1	CCx	Uniform	0.6–0.99	–	–
2020	1	CGC_CD	Uniform	0.08–0.16	–	–
2020	1	CDC_CD	Truncated normal	0.03–0.12	0.07	0.025
2020	1	SenescenceCD	Truncated normal	115–145	132	10
2020	1	Kcb	Uniform	1.05–1.6	–	–
2020	2	WP_early	Uniform	10–40	–	–
2020	2	WP_late	Uniform	10–40	–	–

Year	Phase	Parameter	Family	Support	Center	Scale
2020	2	HI0	Uniform	0.35–0.6	–	–
2021	1	CCx	Uniform	0.6–0.99	–	–
2021	1	CGC_CD	Uniform	0.08–0.16	–	–
2021	1	CDC_CD	Truncated normal	0.03–0.12	0.07	0.025
2021	1	SenescenceCD	Truncated normal	115–145	132	10
2021	1	Kcb	Uniform	1.05–1.6	–	–
2021	2	WP_early	Uniform	10–40	–	–
2021	2	WP_late	Uniform	10–40	–	–
2021	2	HI0	Uniform	0.35–0.6	–	–
2022	1	CCx	Truncated normal	0.6–0.99	0.988869	0.05
2022	1	CGC_CD	Truncated normal	0.08–0.16	0.104272	0.015
2022	1	CDC_CD	Truncated normal	0.03–0.12	0.0688532	0.025
2022	1	SenescenceCD	Truncated normal	115–145	133	10
2022	1	Kcb	Truncated normal	1.05–1.6	1.42159	0.1
2022	2	WP_early	Truncated normal	10–22	16.0015	3
2022	2	WP_late	Truncated normal	20–38	29.5521	4
2022	2	HI0	Truncated normal	0.35–0.5	0.4	0.03

Note. The common independent Sobol support is a separate experiment from these annual MCMC priors.

S3 原链诊断 / Original-chain diagnostics

Table S3. Time-preserving diagnostics and retained-mean sensitivity.

Year	Phase	Independent ensembles	Worst $\hat{R}$	Min group bulk ESS	Max τ	Burn-in shift (SD)
2019	1	4	1.1806	17.79	78.81	0.0327
2019	2	1	–	–	34.45	0.0305
2020	1	4	1.1739	19.18	73.77	0.0168
2020	2	1	–	–	52.34	0.0246
2021	1	4	1.1239	24.94	69.37	0.0362
2021	2	1	–	–	47.76	0.0176
2022	1	4	1.1222	25.34	63.40	0.0395
2022	2	1	–	–	32.58	0.0314

Note. Stage-1 extrema summarize matched walker-index comparisons across four independently initialized ensembles while preserving time. Dependent groups are not added to form a pooled ESS. Stage 2 has one independent ensemble per year.

四年阶段一保留均值对截取方式较稳定，分布采样精度由 $\hat{R}$、ESS 和 τ 另行量化。2021 年短时续链记录独立归档：公共链长 2532 步时最差 $\hat{R}$为 1.0888、最小组 bulk ESS 为 33.64。该补充计算未替换主分析参数中心。

Retained means and distributional precision are evaluated separately. A separately archived 2021 continuation has a common length of 2532 steps, worst $\hat{R}$=1.0888 and minimum matched-group bulk ESS=33.64; it does not replace the four-year parameter centers used here.

S4 五期日期界限 / Five-stage date bounds

Table S4. Eligible dates for all five stages in all years.

Year	Stage	First date	Last date	Max events
2019	Seedling	2019-04-18	2019-05-31	3
2019	Jointing	2019-06-01	2019-07-09	3
2019	Heading	2019-07-10	2019-08-04	3
2019	Grain filling	2019-08-05	2019-08-31	3
2019	Maturity	2019-09-01	2019-09-15	3
2020	Seedling	2020-04-20	2020-06-06	3
2020	Jointing	2020-06-07	2020-07-13	3
2020	Heading	2020-07-14	2020-07-27	3
2020	Grain filling	2020-07-28	2020-08-20	3
2020	Maturity	2020-08-21	2020-09-18	3
2021	Seedling	2021-04-27	2021-06-11	3
2021	Jointing	2021-06-12	2021-07-12	3
2021	Heading	2021-07-13	2021-07-28	3
2021	Grain filling	2021-07-29	2021-08-20	3
2021	Maturity	2021-08-21	2021-09-21	3
2022	Seedling	2022-04-17	2022-06-01	3
2022	Jointing	2022-06-02	2022-07-04	3
2022	Heading	2022-07-05	2022-07-20	3
2022	Grain filling	2022-07-21	2022-08-12	3
2022	Maturity	2022-08-13	2022-09-11	3

Note. Dates follow the reported phenological observations [1], intersected with the executed model calendar. Every stage is eligible even where the recorded schedule has no event.

S5 2019 年逐次日程 / Individual events in 2019

Table S5. Every positive-depth event in the selected 2019 schedule.

Stage	Date	Model day (1-based)	Actual depth (mm)	Reading
Seedling	2019-05-06	19	81.720754	≥0.1 mm
Seedling	2019-05-07	20	3.447512	≥0.1 mm
Jointing	2019-06-13	57	0.220281	≥0.1 mm
Jointing	2019-06-14	58	4.224564	≥0.1 mm
Jointing	2019-06-20	64	0.200626	≥0.1 mm
Heading	2019-07-18	92	117.028543	≥0.1 mm
Heading	2019-07-26	100	2.723275	≥0.1 mm
Grain filling	2019-08-05	110	0.054565	Trace <0.1 mm
Grain filling	2019-08-07	112	0.173138	≥0.1 mm
Grain filling	2019-08-24	129	0.013362	Trace <0.1 mm

Stage	Date	Model day (1-based)	Actual depth (mm)	Reading
Maturity	2019-09-01	137	0.024961	Trace <0.1 mm
Maturity	2019-09-03	139	1.16e-09	Trace <0.1 mm

Note. Model day 1 is the planting date; calendar dates are explicit. Numerical traces are retained without changing the evaluated controls. Zero slots are available in the complete source table.

S6 2020 年逐次日程 / Individual events in 2020

Table S6. Every positive-depth event in the selected 2020 schedule.

Stage	Date	Model day (1-based)	Actual depth (mm)	Reading
Seedling	2020-04-20	1	108.287640	≥0.1 mm
Jointing	2020-06-10	52	0.174299	≥0.1 mm
Jointing	2020-06-21	63	93.930726	≥0.1 mm
Jointing	2020-07-01	73	0.599161	≥0.1 mm
Heading	2020-07-16	88	2.160578	≥0.1 mm
Heading	2020-07-25	97	0.092987	Trace <0.1 mm
Heading	2020-07-26	98	0.001515	Trace <0.1 mm
Grain filling	2020-07-28	100	87.916543	≥0.1 mm
Grain filling	2020-07-29	101	0.790977	≥0.1 mm
Grain filling	2020-08-17	120	0.003147	Trace <0.1 mm
Maturity	2020-08-23	126	12.465925	≥0.1 mm
Maturity	2020-09-02	136	0.034240	Trace <0.1 mm
Maturity	2020-09-09	143	0.041215	Trace <0.1 mm

Note. Model day 1 is the planting date; calendar dates are explicit. Numerical traces are retained without changing the evaluated controls. Zero slots are available in the complete source table.

S7 2021 年逐次日程 / Individual events in 2021

Table S7. Every positive-depth event in the selected 2021 schedule.

Stage	Date	Model day (1-based)	Actual depth (mm)	Reading
Seedling	2021-04-27	1	0.000663	Trace <0.1 mm
Seedling	2021-05-23	27	0.002344	Trace <0.1 mm
Jointing	2021-06-15	50	92.959220	≥0.1 mm
Jointing	2021-06-28	63	85.822216	≥0.1 mm
Jointing	2021-07-09	74	0.000563	Trace <0.1 mm
Heading	2021-07-20	85	0.053827	Trace <0.1 mm
Grain filling	2021-07-29	94	85.634129	≥0.1 mm
Maturity	2021-08-21	117	0.008609	Trace <0.1 mm
Maturity	2021-08-26	122	0.018403	Trace <0.1 mm

Note. Model day 1 is the planting date; calendar dates are explicit. Numerical traces are retained without changing the evaluated controls. Zero slots are available in the complete source table.

S8 2022 年逐次日程 / Individual events in 2022

Table S8. Every positive-depth event in the selected 2022 schedule.

Stage	Date	Model day (1-based)	Actual depth (mm)	Reading
Seedling	2022-04-17	1	0.035146	Trace <0.1 mm
Jointing	2022-06-02	47	0.109382	≥0.1 mm
Jointing	2022-06-11	56	181.310872	≥0.1 mm
Heading	2022-07-08	83	2.460366	≥0.1 mm
Heading	2022-07-18	93	0.474977	≥0.1 mm
Grain filling	2022-07-21	96	71.524972	≥0.1 mm
Grain filling	2022-07-22	97	1.505109	≥0.1 mm
Maturity	2022-08-13	119	0.044430	Trace <0.1 mm

Note. Model day 1 is the planting date; calendar dates are explicit. Numerical traces are retained without changing the evaluated controls. Zero slots are available in the complete source table.

S9 单次量与实施条件 / Event magnitude and implementation conditions

Table S9. Maximum event depths and spacing.

Year	Recorded max (mm)	Selected max (mm)	Events ≥0.1 mm	Min positive-event gap (d)
2019	67.71	117.03	8	1
2020	84.78	108.29	8	1
2021	114.80	92.96	3	5
2022	85.85	181.31	6	1

Note. No additional equipment cap, minimum-event-depth constraint, spacing constraint or terminal-water floor is imposed. Gaps include exact trace events.

S10 原控制空间补充试验 / Earlier control-space diagnostics

前期四窗口单目标求解、2021 年 1 m 根深与不同 fMulch 条件重校准，以及降水替换试验均保持原始结果。相应试验分别回答固定日期空间的算法重复性、条件结构改变及降水来源敏感性，主文五期日程采用本轮新搜索。完整源码、响应和图件见归档 05—10 及 15—16 阶段。

Earlier scalar comparisons retain four fixed-date windows. The 2021 1 m root-depth/mulch conditional refits and rainfall substitutions retain their own settings. Their results answer distinct control-space, structural and forcing questions; current five-stage calendars use the new searches. Complete source code and results are indexed by research stage.

Table S10. Earlier scalar-search archive minima in the corresponding fixed-date spaces.

Case	Method	I min (mm)	Yield (t ha^{-1})
old_2019	NSGA2_1obj	217.95	15.723
old_2020	NSGA2_1obj	313.01	16.430
old_2021	GA	268.61	15.067
old_2022	NSGA2_1obj	261.46	13.377
new_fm050	GA	316.48	15.070
new_fm075	GSA	280.15	15.067
new_fm085	NSGA2_1obj	286.86	15.072

Note. These are earlier control-space diagnostics; the present 30D comparison is in main Table 10. Historical NSGA-III reference budgets differ from the scalar-query budgets.

S11 数值复现 / Numerical reproduction

The search source is pinned to Git commit 20f1616556f5aeb79329b7ead22faa28f9fcda6a. All 457,596 query rows, date bounds, per-stage limits and seasonal budgets were checked. Full-model selected-point replays and 12 fresh unaccelerated original-equation replays agree within 1.14×10^{-13}. Source JSON records separately count exact cache hits, setup, preflight and postprocessing calls. Four first seeds retain 9,999 queries after duplicate initialization; no synthetic rows were added. This manuscript update performs no new MCMC, Sobol or optimization runs.